Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.

Experiments on two machine translation tasks show the Transformer is more parallelizable and requires significantly less time to train. The Transformer achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the best previously reported results.

On the WMT 2014 English-to-French translation task, the big Transformer model establishes a single-model state-of-the-art BLEU score of 41.8.

Scaled dot-product attention computes Attention(Q, K, V) = softmax(QK^T / sqrt(d_k)) V. We use h = 8 parallel attention heads in the multi-head attention layer.

References

[1] Dzmitry Bahdanau et al. Neural Machine Translation by Jointly Learning to Align and Translate. ICLR 2015.